A-A-53856
15 March 1989

COMMERCIAL ITEM DESCRIPTION

REFRACTOMETER

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-R-37053B which will be cancelled.

This commercial item description covers a lightweight, hand-held, monocular refractometer used to measure specific gravity of human urine, serum protein concentration, and refraction.

Salient characteristics.

The refractometer shall be a lightweight, hand-held, monocular optical instrument designed to measure refractive index. It shall be calibrated to provide scales for direct reading of specific gravity of human urine, serum protein concentration, and refraction, on a graduated transparent reticle through a focusable, recessed eyepiece. Shall be a transmission type designed to provide maximum contrast between dark and light. Shall be designed to automatically compensate for temperature variations.

Shall be provided with means for recalibration, and shall be contained in a cushioned protective case. Weight of the unit shall be not more than 11 ounces.

The refractometer shall consist of a tubular shaped body containing an eyepiece, a reticle, an objective, a measuring prism, a prism cover, a movable cover plate, and a case.

The body shall be constructed of 2011 aluminum with a clear anodized finish. Overall dimensions shall be 6-3/8 $\pm$ 1/4 inches in length by 1-1/4 $\pm$ 1/8 inches in diameter.

AMSC N/A FSC 6650

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The eyepiece at the proximal end of the instrument shall be a lens having a magnification of 10X. The lens element shall be made of clear optical glass and shall be low-reflection coated. The eyepiece shall be focusable to compensate for differences in eyesight characteristics, and shall magnify the scales on the reticle for ease of readability.

The reticle shall consist of a magnifying glass lens containing three direct reading scales, one for urine specific gravity, one for serum protein concentration in grams/100 milliliters, and one for refraction. With sample specimen in place, the optical system shall produce a dark area on the reticle. The highest reading on the visible portion (bright area) of the applicable scale on the reticle shall be the quantitative value of the sample specimen.

The scales shall be effected on the reticle lens by a photo etch process with a chrome deposit. Scales shall be clear and distinctly readable when viewed through the eyepiece. Specific gravity scale shall read from 1.000 to 1.035 and shall be graduated in 0.001 intervals, with values marked at each 0.005 graduation. Serum protein scale shall be from 2.5 to 15.0 grams/100 milliliters and shall be graduated in 0.1 grams/100 milliliter intervals with values marked at each 0.5 graduation. Refraction scale shall be from 0 to 320, shall be graduated in increments of 2, and every fifth increment shall be numbered. Scales shall be coordinated as follows:

Refraction	Urine Specific Gravity
0	1.000
60	1.017
	Serum Protein
96	3.9 to 4.0
126	5.5 to 5.6
242	11.0

The objective glass lens, a component of the optical system in the refractometer body tube, shall be designed so as to direct a beam of light in a true path to the scales on the reticle.

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The measuring prism shall consist of an hermetically sealed glass which contains a clear homogenous stable liquid and an air bubble. Shall also include a sealed glass plate prism cover. The index of refraction of the liquid shall change with temperature in a manner which provides automatic compensation for temperature variations. The air bubble shall be of sufficient size to allow for expansion and contraction of liquid due to temperature variation.

The prism cover shall be a framed clear, optical quality glass plate with flat, optically polished, parallel surfaces forming a window at the distal end of the instrument. Overall dimensions shall be 1-9/16 ± 1/16 inches long by 5/8 ± 1/32 inches wide and 1/8 ± 1/32 inches thick. The prism cover shall be the specimen sample holder and window for the beam of light entrained through the optical system.

The cover plate shall be constructed of a clear plastic, 7/32 ± 1/16 inches thick, and of sufficient size to cover the prism cover. Semicircular indentations centrally located on the upper and lower ends of the cover plate shall be provided to facilitate placing sample specimen on the exposed portion of the prism cover so that the liquid is drawn between the two facing surfaces. Coverplate shall be mounted on corrosion-resisting metal wire supports so as to be capable of readily being swung onto, or completely away from the prism cover. In position of use, coverplate and prism cover facing surfaces shall accommodate a thin layer of liquid sample. With the coverplate swung away from the prism cover, the refractometer shall also be able of being used by immersion into sample specimen.

The case shall be a black vinyl sheath type with dark colored urethane foam lining to afford adequate protective cushioning for the instrument. Case shall have a flap-type cover with adequate means for effecting a secure closure. Centrally positioned on the proximal end of the back side of the case shall be a nominal 2-1/4 inch, sturdy, pocket-type clip.

The refractometer shall be provided with means for in-field calibration in the form of a setscrew beneath the objective lens which shall be readily set with a screwdriver. The opening over the setscrew shall be filled with a sealer to prevent leaking of moisture into the instrument.

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The refraction scale shall be tested at zero and at a value above 200. Testing shall be performed by comparison with distilled water and other standards. The refractometer shall be operated in accordance with instructions in service data. Refraction standards shall be verified with a master refractometer at 20° C.

The refractometer shall be permanently and legibly marked with the following information:

National Stock Number
Manufacturer's name or registered trademark
Model Number
"U.S."

The case shall be permanently marked with the following information:

Item Identification
Manufacturer's name or registered trademark

Workmanship. The refractometer shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One refractometer, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Inspections and tests. Inspections and tests shall be conducted in accordance with commercial practice to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for inspections and tests. Sampling for inspections and tests shall be conducted in accordance with commercial practice. The unit of product for tests purposes shall be one

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.refractometer.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U. S. Food and Drug Administration to be under its jurisdiction, the offeror/ contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations

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promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

6650-00-933-3218

Ordering data. (Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order).

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6650-0170

Location: Disk No. 16
CID/REFRACTOMETER/S38

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
-A-53856	REFACTOMETER

NAME OF SUBMITTING ORGANIZATION

b ADDRESS (*Street City State ZIP Code*)

4 TYPE OF ORGANIZATION (*Mark one*)

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER (*Specify*)

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER (*Last First MI*) – Optional

b WORK TELEPHONE NUMBER (*Include Area Code*) – Optional

c MAILING ADDRESS (*Street City State ZIP Code*) – Optional

8 DATE OF SUBMISSION (*YYMMDD*)

DD FORM 82 MAR 1426

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